

A Discrete-Sine-Transform based Preconditioner for Adaptive Meshes

Kate Wall

September 15, 2026

In collaboration with: James Adler, Xiaozhe Hu, Misha Kilmer

Abstract

We present a preconditioner for solving fractional differential equations (FDEs) on an adaptive mesh. Adaptive refinement of the problem domain results in a stiffness matrix with Toeplitz blocks along the main diagonal, while the FDE yields a dense stiffness matrix where off-diagonal blocks are stored as low-rank approximations. Our adaptive preconditioner utilizes ideas from the circulant preconditioner of Chan and Strang [SIAM Journal on Scientific Computing, 1989] and the discrete-sine-transform based preconditioner of Bini and Di Benedetto [Association for Computing Machinery, 1990]. We take advantage of the Toeplitz blocks on the diagonal and accounts for the low-rank nature of the off-diagonal blocks. We demonstrate the adaptive preconditioner's effectiveness at accelerating convergence for our systems and emphasize its efficient application. This work presents theoretical results about the spectral clustering of the preconditioned system. In order to prove these results, special consideration is taken on how the low-rank blocks perturb the eigenvalues of the Toeplitz block-diagonal system. Numerical tests are used to inspect any assumptions and validate our results.

1 Introduction

In this paper we consider fractional differential equations (FDEs) of the form

$$D_x^\alpha u(x) = f(x), \quad x \in (a, b), \quad (1)$$

where D_x^α is the Riesz fractional derivative of order $1 < \alpha < 2$ and $f \in L^2([a, b])$.

Fractional equations find applications in many fields such as fluid mechanics, electromagnetic fluids, image processing, and material science (see for example [13, 8]). However, solving FDEs presents unique difficulties in numerical

methods due to the non-locality of the operator resulting in a dense stiffness matrix, A . Additionally, as part of the fractional derivative the singular kernel, $k(x, s) = |x - s|^{1-\alpha}$, is integrated, resulting in end-point singularities which make Galerkin finite-element methods sub-optimal [11, 16]. To address this, Jin and Zhou developed a reconstruction technique that isolates the singularity to improve finite-element method (FEM) convergence [9]. Alternatively, Wang and Du developed a Toeplitz preconditioner that reduces iteration cost to $\mathcal{O}(m \log m)$, where m is the size of A , using the preconditioned conjugate gradient squared method [16].

The above methods require uniform meshes. On a uniform mesh the FEM discretization results in a stiffness matrix that is Toeplitz. All methods on a uniform mesh leverage this structure for efficient linear solves. However, these methods fail to address high error near singularities, and instead the size of the mesh must grow. An optimal mesh would have the L^∞ error equally distributed on each interval, but a uniform mesh does not, in general, satisfy this condition. Indeed, the singularities of FDEs force higher error in the intervals near the boundary and thus require very fine meshes to achieve uniform error below a given threshold. Alternatively, to achieve desired accuracy across the whole domain we use an adaptive mesh and adaptive finite-element method (AFEM) [12, 17]. The usual AFEM algorithm iterates through the process

$$\text{SOLVE} \rightarrow \text{ESTIMATE} \rightarrow \text{MARK} \rightarrow \text{REFINE}, \quad (2)$$

but the computational cost is dominated by the **SOLVE** step.

Zhao et al. used AFEM to solved FDEs using hierarchical matrices and geometric multigrid (GMG) [17]. This method is done in $\mathcal{O}(kN \log N)$ where k is a parameter for low-rank approximations and N is the degrees of freedom at the finest level. Numerical tests show this technique attains optimal convergence order, but there are no proofs guaranteeing such.

In this paper, we construct a preconditioner, M , for the stiffness matrix, A , from an AFEM discretization of Equation (1). We show that the cost of solving the preconditioned linear system with a Krylov method is competitive with the GMG approach and furthermore prove that the preconditioned spectrum is clustered around 1, guaranteeing convergence. Our approach also has the advantage of avoiding hierarchical matrices and their sometimes cumbersome implementation.

In the next section we present background information about structured matrices and preconditioning Toeplitz matrices. Then in Section (3) we investigate what structure the adaptive mesh retains and how that is reflected in the structure of A . We leverage this structure to apply preconditioning techniques formulated for Toeplitz matrices [4, 5, 3, 7]. This is followed by theoretical results about eigenvalue clustering of the preconditioned system in Section(4). Numerical results to validate any of our assumptions about the stiffness matrix and compare performance to the Zhao et al.'s GMG method are presented in Section (5).

2 Background

2.1 Properties of Structured Matrices

There has long been interest in solving Toeplitz linear systems efficiently. A matrix A is called Toeplitz if $a_{ij} = a_{i-j}$. Arbitrary $n \times n$ matrices have up to n^2 entries and are solved directly in $\mathcal{O}(n^3)$ FLOPS. Since a Toeplitz matrix has at most $2n - 1$ unique entries, we expect to solve Toeplitz systems with $\mathcal{O}(n^2)$ FLOPS, (see for example Levinson's algorithm, Trench's algorithm). Even with these improvements, however, fast Toeplitz algorithms are unstable in general and infeasible for large systems that arise in application [2]. Infinite or periodic Toeplitz matrices can be solved via a transformation to Fourier space and a discrete convolution in $\mathcal{O}(n)$ time. On the other hand, finite Toeplitz systems are much more difficult to solve due to the boundaries of the mesh or matrix. To solve these systems, we turn to iterative Krylov and multigrid methods [7, 11, 17]. For these methods we take advantage of Toeplitz structure by using structured preconditioners.

Toeplitz matrices commonly arise in partial differential equations discretization, FDEs, signal processing, and control theory. In application, Toeplitz matrices are often symmetric positive definite (SPD) or symmetric positive semidefinite (SPSD). Given an SPD Toeplitz system $Lx = b$, the idea introduced by Strang in [15] is to use a circulant preconditioner, C , so that $C^{-1}Tx = C^{-1}b$ is solved in fewer iterations. A circulant matrix is a Toeplitz matrix, that additionally has the "wrap-around" property where the last entry of each row is the first entry of the subsequent row:

$$L = \begin{pmatrix} \ell_0 & \ell_{-1} & \cdots & \ell_{-(n-1)} \\ \ell_1 & \ell_0 & \cdots & \ell_{-(n-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n-1} & \ell_{n-2} & \cdots & \ell_0 \end{pmatrix} \quad C = \begin{pmatrix} c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{pmatrix}. \quad (3)$$

Equation 1: L is a Toeplitz matrix and C is circulant.

As will be discussed in Section (2.2), the stiffness matrix resulting from AFEM applied to (1) is not Toeplitz, but does contain SPD Toeplitz blocks. Toeplitz matrices have many special properties, here we focus on their connection to functions in the Wiener class.

Definition 1 ([4]). *Given a singly infinite, symmetric Toeplitz matrix L we*

define the finite subsection L_m to be the $m \times m$ principle submatrix. That is,

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & & \\ \ell_1 & \ell_0 & \ell_1 & \ddots & \\ \ell_2 & \ell_1 & \ell_0 & \ddots & \\ & \ddots & \ddots & \ddots & \ddots \end{pmatrix} \quad L_m = \begin{pmatrix} \ell_0 & \ell_1 & \cdots & \ell_{m-1} \\ \ell_1 & \ell_0 & \cdots & \ell_{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{m-1} & \ell_{m-2} & \cdots & \ell_0 \end{pmatrix}.$$

Assume $\sum_{k=-\infty}^{\infty} |\ell_k| < \infty$, then the matrices induce functions $\tilde{\ell}(z) = \sum_{k=-\infty}^{\infty} \ell_k z^k$ and $\tilde{\ell}_n(z) = \sum_{k=-(n-1)}^{n-1} \ell_k z^k$ are real, positive, and in the Wiener class for $|z| = 1$.

Previous work on Toeplitz systems offers many options for circulant preconditioners, (see [3]). Strang and R. Chan first proposed to construct a circulant C by copying the central diagonals of L into the required wrap-around entries. Assuming for simplicity $n = 2m$, then the first column of C would be $\ell_0, \dots, \ell_m, \dots, \ell_0$ [4]. Alternatively, Tony Chan constructed a circulant preconditioner, C , by averaging the diagonals of L . He shows this to be optimal in the sense that $\|C - L\|_F$ is minimal [6].

Although their construction differs, they are all diagonalized by some trigonometric transform matrix and are thus efficiently applied via the fast Fourier transform (FFT). The spectrum of the preconditioned systems are asymptotically the same [3]. In this paper the preconditioner we construct is based on the φ matrix class, introduced in [1].

The φ preconditioner is not itself circulant, but is diagonalized by the type-I discrete sine transform (DST) matrix. Kailath and Olshevsky used a displacement operator kernel framework to unify the circulant preconditioners, φ , and other preconditioners diagonalized by other discrete trigonometric matrices. Together these form a class of discrete-trigonometric-based preconditioners. Though our construction could be generalized to circulant preconditioners, we use φ because it does not require complex arithmetic needed for the FFT.

Definition 2. Following [1], given a symmetric $n \times n$ Toeplitz matrix L , we

build a Hankel correction matrix, H , where

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \cdots & \ell_{n-2} & \ell_{n-1} \\ \ell_1 & \ell_0 & \ell_1 & \cdots & \ell_{n-3} & \ell_{n-2} \\ \ell_2 & \ell_1 & \ell_0 & \cdots & \ell_{n-4} & \ell_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \ell_{n-2} & \ell_{n-3} & \ell_{n-4} & \cdots & \ell_0 & \ell_1 \\ \ell_{n-1} & \ell_{n-2} & \ell_{n-3} & \cdots & \ell_1 & \ell_0 \end{pmatrix}$$

$$H = \begin{pmatrix} \ell_2 & \ell_3 & \ell_4 & \cdots & \ell_{n-1} & 0 & 0 \\ \ell_3 & \ell_4 & \ell_5 & \cdots & 0 & 0 & 0 \\ \ell_4 & \ell_5 & \ell_6 & \cdots & 0 & 0 & \ell_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \ell_5 & \ell_4 & \ell_3 \\ 0 & 0 & \ell_{n-1} & \cdots & \ell_4 & \ell_3 & \ell_2 \end{pmatrix}.$$

We can additionally define $\varphi := L - H$.

Both L and H are symmetric, so they are completely represented by at most n unique values: the diagonals of $L : \ell_0, \ell_1, \dots, \ell_{n-1}$, and the anti-diagonals of $H : \ell_2, \ell_3, \dots, \ell_{n-1}, 0, 0$.

The following lemma summarizes properties of φ .

Lemma 1 (see [1]). *Let L be a Toeplitz matrix and H the corresponding Hankel correction as in Definition (2). Then $\varphi = L - H$ has the following properties:*

1. φ is diagonalized by the DST, S . We write $\varphi = SAS^{-1} = SAS$.
2. φ can be applied in $\mathcal{O}(n \log n)$ FLOPS.
3. Let L_1 be the first column of L . Then, the k -th eigenvalue of φ is proportional to the k -th entry of SL_1 . Specifically,

$$\lambda_k(\varphi) = c_k [SL_1]_k, \text{ where } c_k := \sqrt{\frac{n+1}{2}} \frac{1}{\sin(\frac{\pi k}{n+1})}. \quad (4)$$

4. Each eigenvalue of φ is written as the truncated function $\tilde{\ell}_n$ evaluated somewhere on the unit circle, $\lambda_k(\varphi) = \tilde{\ell}_n(z_k)$, where $|z_k| = 1$.

2.2 AFEM stiffness matrix

Again, let A be the stiffness matrix resulting from AFEM applied to (1). During the MARK module of AFEM (2), it is often the case that adjacent elements have similar error estimates, and are either both marked or both not marked for refinement. As a result, after several refinements there are still subsections of the domain that have been refined to the same level. We refer to these subdomains as *locally uniform*. Since uniform meshes give rise to SPD Toeplitz systems,

wherever there are adjacent elements of the same size we find a corresponding Toeplitz block on the main diagonal of our stiffness matrix, A . The size of this block depends on how many adjacent elements are the same size. In practice we have observed the greatest level of refinement is required near the boundary of the domain. In general we have larger Toeplitz blocks from locally uniform subdomains in the middle of the matrix, and smaller blocks—or indeed 1×1 blocks—near the boundary.

These 1×1 blocks, along with their corresponding rows and columns, can be viewed as additional boundaries imposed onto the system. As mentioned earlier, it is the boundaries of a Toeplitz system that necessitate more computation while solving, so we can anticipate that wise handling of these 1×1 block disruptions to the Toeplitz structure will be the key to making a good preconditioner. In order to better handle these 1×1 blocks, we solve the a permuted system, $\tilde{A}$, shifting the rows and columns associated with these blocks to the upper left corner of the matrix, constituting V, W and $W^\top$, as shown in Figure (1). Thus along the diagonal of the lower right block, T , remain Toeplitz blocks of size at least two.

$$PAP^\top = \tilde{A} = \begin{bmatrix} V & W \\ W^\top & T \end{bmatrix}$$

Figure 1: The stiffness matrix A is permuted and partitioned so that T contains all Toeplitz blocks along its diagonal.

Let k denote the size of V , that is the number of 1×1 blocks. It is natural to ask, and will indeed be important in constructing a preconditioner, how big is k relative to $m = \text{size}(A)$? It is difficult to get an exact handle on this theoretically, since it depends on how AFEM routines are implemented. However, in our numerical experiments we found that $k \approx \log(m)$ (see Section (5.1)), and k never grew faster than m .

We wish to further break down the structure of T . Due to the permutation, T has Toeplitz blocks of size at least 2 along its diagonal. We use T_D to denote the block diagonal with Toeplitz blocks.

The fractional derivative results in a dense matrix, but we observe off-diagonal decay, as elements that are physically separated in the domain have weaker interaction. The rate of decay depends on the fractional order α . Blocks that are most distant from the diagonal contain less information and we assume they have small-norm. We denote these blocks in T as T_F . We call the blocks that are either directly below or to the right of the Toeplitz blocks “blocks adjacent to the diagonal”, together they are denoted T_E . These blocks are not in general small-norm, but due to the weaker interaction, are well-suited for low-rank approximation, (see [17] for more details). These assumptions are made precise in section 4.2 and numerically validated in section 5.1. Using this notation, we write our stiffness matrix as the sum of the Toeplitz blocks, blocks adjacent to the diagonal, and blocks further removed from the diagonal,

$T = T_D + T_E + T_F$ (see Figure (2)).

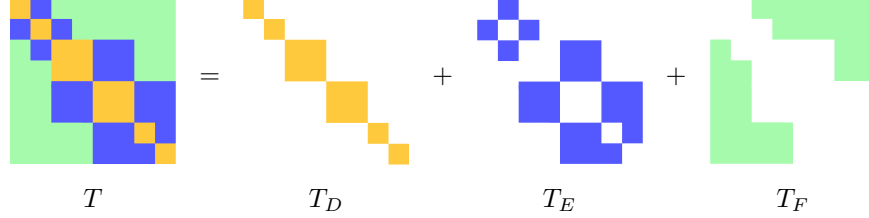


Figure 2: Illustration of the splitting of T : T_D is block diagonal with Toeplitz blocks, blocks in T_E are assumed to be well-approximated by low-rank matrices, and T_F is assumed to be small-norm due to weak interaction between elements.

3 The Adaptive Preconditioner

In this section we set forth the properties we require from a preconditioner. Using φ for its efficient inversion and good approximation of Toeplitz blocks, we construct our preconditioner, M , as an approximation to the LU decomposition of the permuted stiffness matrix.

3.1 Preconditioner Requirements

A good preconditioner for an iterative method must in general decrease the total number of iterations, without increasing the cost of a single iteration. We borrow Kailath's [10] criteria for preconditioners, similar criteria is well-established in the literature (for example [1]).

1. Complexity of constructing and applying $M \in \mathbb{R}^{m \times m}$ should be $\mathcal{O}(m \log m)$.
2. A linear system with M should be solved in $\mathcal{O}(m \log m)$ operations.
3. The spectrum of $M^{-1}\tilde{A}$ should be clustered around 1.

3.2 Constructing our Preconditioner

3.2.1 LU Decomposition of $\tilde{A}$

To construct our preconditioner we must consider the permuted 1×1 blocks in V and the structured pieces of T . Using the Schur complement, we can obtain a block-LU decomposition for our permuted stiffness matrix. It will be notationally convenient to define the Schur complement matrix $B := T - W^\top V^{-1}W$.

$$\tilde{A} = LU = \begin{pmatrix} I & 0 \\ W^\top V^{-1} & I \end{pmatrix} \begin{pmatrix} V & W \\ 0 & B \end{pmatrix} \quad (5)$$

It is straightforward to verify the inverses of the two factor matrices are,

$$L^{-1} = \begin{pmatrix} I & 0 \\ -W^\top V^{-1} & I \end{pmatrix}$$

$$U^{-1} = \begin{pmatrix} V^{-1} & -V^{-1}WB^{-1} \\ 0 & B^{-1} \end{pmatrix}$$

Recalling that $k = \text{size}(V)$ is much smaller and slower-growing than n , inverting V will not contribute to our overall costs. Next, we can invert B using the Sherman-Woodbury-Morris formula, $B^{-1} = T^{-1} + T^{-1}W^\top(V - WT^{-1}W^\top)^{-1}WT^{-1}$. Notice, however, it requires the prohibitively expensive inversion of T . Here, we must leverage the Toeplitz blocks to build an adequate approximation to T^{-1} .

3.2.2 Approximating T^{-1}

Let $L_1, L_2, \dots, L_p$ be the Toeplitz blocks ordered as they appear along the diagonal of T_D , with respective sizes $m_1, m_2, \dots, m_p$. Then for each L_i we construct φ_i as in Definition (2). Stacking these blocks we build a block φ matrix as $\Phi := \text{diag}(\varphi_1, \varphi_2, \dots, \varphi_p)$. By application of Lemma 1 on each φ_i , we can apply Φ^{-1} implicitly in $\mathcal{O}\left(\sum_{j=1}^k m_j \log(m_j)\right) \leq \mathcal{O}(km \log(m))$, where $m := \max_j m_j$. Let S_{m_i} denote the type-I DST matrix of size m_i . Using Equation (4) to calculate the eigenvalues of φ_i , we write $\varphi_i = S_{m_i} \Lambda_i S_{m_i}$.

$$\Phi^{-1} = \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_k} \end{pmatrix} \begin{pmatrix} \Lambda_1^{-1} & & & \\ & \Lambda_2^{-1} & & \\ & & \ddots & \\ & & & \Lambda_m^{-1} \end{pmatrix} \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_k} \end{pmatrix}. \quad (6)$$

The DST blocks in the outside matrices can be applied via the fast sine transform (FST) to do matrix multiplication in $\mathcal{O}(m \log m)$ FLOPS. The eigenvalue matrix is diagonal and inverted in m operations, and can be multiplied against a Toeplitz matrix embedded in a $2m \times 2m$ circulant matrix in $\mathcal{O}(m \log m)$ FLOPS. Moreover, both the FST and block arithmetic are highly parallelizable for more efficient implementation. This establishes Criteria (1) and (2) for Φ .

Therefore Φ^{-1} is fast to compute, and in Section (4) we will show it is a good spectral approximation to T^{-1} .

3.2.3 Defining the Adaptive Preconditioner, M^{-1}

Let $\hat{B} = \Phi - W^\top V^{-1}W$. Then define

$$\hat{U} := \begin{pmatrix} V & W \\ 0 & \hat{B} \end{pmatrix}, \quad \hat{U}^{-1} = \begin{pmatrix} V^{-1} & -V^{-1}W\hat{B}^{-1} \\ 0 & \hat{B}^{-1} \end{pmatrix}.$$

Our preconditioner is given by

$$\begin{aligned} M^{-1} &:= \hat{U}^{-1} L^{-1} = \begin{pmatrix} V^{-1} & -V^{-1}W\hat{B}^{-1} \\ 0 & \hat{B}^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -W^\top V^{-1} & I \end{pmatrix} \\ &= \begin{pmatrix} V^{-1} + V^{-1}W\hat{B}^{-1}W^\top V^{-1} & -V^{-1}W\hat{B}^{-1} \\ -\hat{B}^{-1}W^\top V^{-1} & \hat{B}^{-1} \end{pmatrix}. \end{aligned}$$

The cost of applying M^{-1} is dominated by application of the biggest block, $\hat{B}^{-1}$. However, by using Φ in place of T , and consequently $\hat{B}$ in place of B , we use the Sherman-Morrison-Woodbury formula to calculate $\hat{B}^{-1} = (\Phi - \tilde{W}^\top \tilde{V}^{-1} \tilde{W})^{-1} = (\Phi^{-1} + \Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1}W\Phi^{-1})^{-1}$. Thus, using that fast application of Φ , as explained above, M^{-1} can be applied in $\mathcal{O}(m \log m)$ FLOPS. Therefore M also satisfies Criteria (1) and (2).

4 Theoretical Results

How tightly the eigenvalues cluster around 1 determines the speed of convergence. In this section we prove that the spectral clustering in criteria (3) holds for M . Proofs for clustering of discrete-trigonometric-transform based preconditioned problems in this area tend to follow a similar structure: show the preconditioned matrix can be written as the sum of the identity matrix, a small-norm matrix, and a low-rank matrix. The small-norm term has spectrum clustered around 0, forcing the clustering of the entire system around 1, with the number of outliers bounded by the rank of the low-rank term.

We build up to the clustering of the final preconditioned system in several steps. First, we briefly present our version of the spectral clustering proof for the case of φ acting on a single Toeplitz block. Then we expand this result to prove the clustering of $\Phi^{-1}T$; that is the block Φ matrix applied to the Toeplitz blocks and corresponding off diagonal blocks of A_{22} . Finally, we show how our handling of the 1×1 blocks preserves the spectral clustering of the whole system, $M^{-1}\tilde{A}$.

The following lemma will be of use to us throughout the section.

Lemma 2. *Let $T \in \mathbb{R}^{n \times n}$ be an invertible matrix. Let $A \in \mathbb{R}^{n \times n}$, then*

$$\lambda_k(T^{-1}A) \leq \frac{\lambda_k(A)}{\lambda_{\min}(T)}.$$

Proof. Using the min-max theorem,

$$\begin{aligned}
\lambda_k(T^{-1}A) &= \min_{\dim V=k} \max_{x \in V} \left(\frac{(Ax, x)}{(Tx, x)} \right) \\
&\leq \min_{\dim V=k} \left[\max_{x \in V} \left(\frac{(Ax, x)}{(x, x)} \right) \max_{x \in V} \left(\frac{(x, x)}{(Tx, x)} \right) \right] \\
&\leq \left[\min_{\dim V=k} \max_{x \in V} \left(\frac{(Ax, x)}{(x, x)} \right) \right] \max_{x \in \mathbb{R}^n} \left(\frac{(x, x)}{(Tx, x)} \right) \\
&= \lambda_k(A) \max_{x \in \mathbb{R}^n} \left(\frac{(x, x)}{(Tx, x)} \right) \leq \lambda_k(A) \frac{1}{\lambda_{\min}(T)}.
\end{aligned}$$

□

4.1 Spectral Clustering of a Toeplitz Block

Proofs demonstrating the eigenvalue clustering of a circulant or trigonometric-based preconditioners on a Toeplitz matrices can follow a few strategies, here we briefly give our version, which utilizes the infinite matrix framework set forth in Section (2.1). Proofs in [1] and [10] follow a similar technique, see [4] for an approach relying on the infinite matrix framework and theory of collectively compact operators. In keeping with these proofs, we assume the generating function $\tilde{\ell}^{(i)}(z)$ induced by each Toeplitz block L_i is in the Wiener class and positive on the unit circle. That is, for all $1 \leq i \leq p$, $\tilde{\ell}^{(i)}(z) = \sum_{j=-\infty}^{\infty} |\ell_j| < \infty$ and $\ell^{(i)}(z) > 0$ for $|z| = 1$.

4.1.1 Approximating T^{-1}

Lemma 3 (Uniform positivity on the unit circle). *Let L be a singly infinite, symmetric, Toeplitz matrix with diagonals $\ell_0, \ell_1, \dots$, and L_m is its $m \times m$ truncation. Assume the generating function $\tilde{\ell}(z)$ is in the Wiener class and positive on the unit circle. Then for $|z| = 1$, $\tilde{\ell}_m(z) > \varepsilon$.*

Proof. Since $\tilde{\ell}(z) > 0$ for $|z| = 1$, compactness of the unit circle implies that for some $\varepsilon > 0$, $\tilde{\ell}(z) > 2\varepsilon$ for $|z| = 1$. By the Wiener class assumption, choose N such that $\sum_{j=N}^{\infty} \ell_j z^j \leq \sum_{j=N}^{\infty} |\ell_j| < \varepsilon$. So for all $m \geq N$,

$$2\varepsilon < \ell(z) = \tilde{\ell}_m(z) + \sum_{j=m}^{\infty} \ell_j z^j < \tilde{\ell}_m(z) + \varepsilon.$$

□

Lemma 4 (Eigenvalue Clustering on a Toeplitz Block, [1, 10]). *Let L , L_m , $\tilde{\ell}$, and $\tilde{\ell}_m$ be as above, and define φ as in 2. Then for m large enough, the spectrum of $\varphi^{-1}L$ is clustered around 1.*

Proof. Since $\varphi = L - H$, $\varphi^{-1}L = \varphi^{-1}(\varphi + H) = I + \varphi^{-1}H$, it suffices to show that the eigenvalues of $\varphi^{-1}H$ are clustered around 0.

We split the Hankel matrix into a low-rank and small-norm piece, $H = H^{LR} + H^{SN}$. Where H^{LR} contains the anti-diagonals $\ell_2, \dots, \ell_N$ with $s := \text{rank}(H^{LR}) \ll m$. The small-norm descriptor is justified for $H^{SN} = H - H^{LR}$, a Hermitian $m \times m$ matrix with at most two copies of $\ell_N, \dots, \ell_{m-1}$ in each row and column. Thus $\|H^{SN}\|_2 = \sqrt{\|H^{SN}\|_1 \|H^{SN}\|_\infty} = \|H^{SN}\|_1 < 2\varepsilon$. Hence by Weyl's Inequality, the eigenvalues of H are clustered within 2ε of zero with at most s outliers.

Now by Lemma (2),

$$\lambda_k(\varphi^{-1}H) \leq \lambda_k(H) \frac{1}{\lambda_{\min}(\varphi)} = \lambda_k(H) \frac{1}{\tilde{\ell}_m(z_{\min})} \leq \lambda_k(H) \frac{1}{\varepsilon}. \quad (7)$$

Thus, the clustering of the spectrum of H around 0 implies the same for $\varphi^{-1}H$. $\square$

Equation (7) shows that the bigger $\lambda_{\min}(\varphi)$, the stronger the bound. We can calculate this eigenvalue exactly using Lemma (1), and equation (4). It can also be bounded in terms independent of φ , using the generating function $\tilde{\ell}(z)$, as above.

4.2 Spectral Clustering of the Block System

We now prove the spectral clustering of $\Phi^{-1}T = \Phi^{-1}(T_D + T_E + T_F)$. It is easy to generalize the lemmas above to the block diagonal Φ^{-1} acting only on the Toeplitz blocks, T_D . The work in this section is in showing that $\Phi^{-1}(T_E + T_F)$ can also be split small-norm and low-rank pieces.

Lemma 5. *Let A be the stiffness matrix as described in Section (2.2) and Φ as described in Section (??). Then the spectrum of $\Phi^{-1}T$ is clustered around 1.*

Proof. Recall that the T block of the permuted stiffness matrix is dense with Toeplitz blocks on the diagonal (denoted T_D), low-rank blocks adjacent to the diagonal (denoted T_E), and small-norm blocks elsewhere (denoted T_F), so that $T = T_D + T_E + T_F$. Let H be the block diagonal Hankel correction matrix, observing then that $T_D = \Phi + H$ we see that $T = \Phi + H + T_E + T_F$. Our approach is to split $H + T_E + T_F$ into a small-norm term T^{SN} and a low-rank term T^{LR} . Then

$$\Phi^{-1}T = \Phi^{-1}(\Phi + T^{SN} + T^{LR}) = I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR}. \quad (8)$$

Then it will suffice to show that the spectrum of $\Phi^{-1}T^{SN}$ is clustered around 0 and that $\Phi^{-1}T^{LR}$ contributes a bounded number of eigenvalue outliers.

Splitting H is done via application of the splitting in the proof of Lemma (5) to each block. As before, we have $\varepsilon > 0$ and choose N_i smaller than each m_i so that for every Toeplitz block the truncated function satisfies $a_{m_i}(z) > \varepsilon$ on the unit circle and $\sum_{i=N_i}^{\infty} |a_i| < \varepsilon$. Then $H = H^{LR} + H^{SN}$, by separating

into H^{LR} the anti-diagonals from block i with coefficients $a_0, \dots, a_{N_i}$ and into H^{SN} the anti-diagonals comprising of $a_{N_i+1}, \dots, a_{m_i}$ for each block. We note two important bounds: $\|H^{SN}\|_2 < \varepsilon$ and $\text{rank}(H^{LR}) \leq 2p(N-1)$ where p is the number of Toeplitz blocks and $N := \max_i N_i$.

We now form a splitting of

$$A_E = \begin{pmatrix} 0 & E_1^\top & & & \\ E_1 & 0 & E_2^\top & & \\ & \ddots & \ddots & \ddots & \\ & & E_{p-2} & 0 & E_{p-1}^\top \\ & & & E_{p-1} & 0 \end{pmatrix}.$$

Recall that each block, E_i , is well-approximated as a low-rank matrix. So for $1 \leq i \leq p-1$, we can write $E_i = E_i^{LR} + E_i^{SN}$ such that $\|E_i^{SN}\|_2 < \varepsilon$ and $r_i := \text{rank}(E_i)$. To use these block splittings to decompose T_E , we split each E_i in place and collect the small-norm terms into T_E^{SN} and similarly form T_E^{LR} with the E_i^{LR} blocks in place. Observe that $\|T_E^{SN}\|_2 \leq 2\varepsilon$ and $\text{rank}(T_E^{LR}) \leq (p-1)r$ where $r := \max_i r_i$.

Finally, we make the splitting $T = \Phi + T^{SN} + T^{LR}$, where

$$T^{SN} = H^{SN} + T_E^{SN} + A_F, \quad T^{LR} = H^{LR} + T_E^{LR}.$$

So $\Phi^{-1}T = I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR}$. Now $s := \text{rank}(\Phi^{-1}T^{LR}) \leq \text{rank}(T^{LR}) \leq 2p(N-1) + (p-1)r$ bounds the number of outlier eigenvalues. It remains to show $\Phi^{-1}T^{SN}$ is indeed small norm.

$$\|\Phi^{-1}T^{SN}\|_2 \leq \|\Phi^{-1}H^{SN}\|_2 + \|\Phi^{-1}T_E^{SN}\|_2 + \|\Phi^{-1}T_F\|_2.$$

Since the block multiplication $\Phi^{-1}H^{SN}$ is each φ acting on its corresponding Hankel correction, it is bounded by Lemma (4). Next,

$$\|\Phi^{-1}T_F\|_2 \leq \|\Phi^{-1}\|_2 \|T_F\|_2 = \sigma_{\max}(\Phi^{-1}) \sigma_{\max}(T_F) = \frac{\sigma_{\max}(T_F)}{\lambda_{\min}(\Phi)}.$$

Since T_F are small-norm this value is small. Using Lemma (2) we bound $\|\Phi^{-1}T_E^{SN}\|_2$:

$$\lambda_k(\Phi^{-1}T_E^{SN}) \leq \frac{\lambda_k(T_E^{SN})}{\lambda_{\min}(\Phi)} = \lambda_k(T_E^{SN}) \min_i \left(\tilde{\ell}_i(z_{\min}) \right) \leq \lambda_k(T_E^{SN}) \frac{1}{\varepsilon}. \quad (9)$$

To finish the proof, we utilize the fact that the matrices are SPD and relate the k -th eigenvalue of the preconditioned matrices to their 2-norm,

$$\begin{aligned} \|\Phi^{-1}T^{SN}\|_2 &\leq \|\Phi^{-1}H^{SN}\|_2 + \|\Phi^{-1}T_E^{SN}\|_2 + \|\Phi^{-1}T_F\|_2 \\ &= \lambda_{\max}(\Phi^{-1}H^{SN}) + \lambda_{\max}(\Phi^{-1}T_E^{SN}) + \lambda_{\max}(\Phi^{-1}T_F) \\ &\leq \frac{1}{\lambda_{\min}(\Phi)} (\lambda_{\max}(H^{SN}) + \lambda_{\max}(T_E^{SN}) + \lambda_{\max}(T_F)). \end{aligned}$$

□

The strength of this bound depends on $\lambda_{\min}(\Phi)$, but can be written independently of Φ in terms of the generating functions as in Equation (9). This dependence is the same for the single block case, and appears in similar proofs for discrete-trigonometric preconditioners (see for example [10]).

In practice the block size, m_i , is determined by the AFEM algorithm, which in turn dictates the actual size ε and hence the strength of clustering. Thus bigger Toeplitz blocks in A result in tighter eigenvalue clustering.

Notice that extending the proof from one Toeplitz block to the adaptive stiffness matrix relies only on being able to split the block into a small-norm and a low-rank piece. Thus, nothing about this proof is overly-specific to the φ preconditioner and the methods could easily be extended to the other types of circulant preconditioners given in [4, 5].

4.3 Spectral Clustering of the Preconditioned System

Theorem 1. *Following the established notation, assume $\Phi^{-1}T^{SN} < \varepsilon$ and define $s := \text{rank}(\Phi^{-1}T^{LR})$. Then the spectrum of $M^{-1}A$ is clustered around 1, with at most s outliers.*

Proof. Let B and $\hat{B}$ be defined as in Section (3.2.3). By matrix multiplication and some simplification we get that the preconditioned system has the form:

$$\begin{aligned} M^{-1}\tilde{A} &= \begin{pmatrix} V^{-1} + V^{-1}W\hat{B}^{-1}W^{\top}V^{-1} & -V^{-1}W\hat{B}^{-1} \\ -\hat{B}^{-1}W^{\top}V^{-1} & \hat{B}^{-1} \end{pmatrix} \begin{pmatrix} V & W \\ W^{\top} & T \end{pmatrix} \\ &= \begin{pmatrix} I & V^{-1}W\hat{B}^{-1}(\Phi - T) \\ 0 & \hat{B}^{-1}(T - W^{\top}V^{-1}W) \end{pmatrix}. \end{aligned}$$

Since this is an upper-triangular system with a $k \times k$ identity block on the diagonal, we know that 1 is an eigenvalue of the preconditioned system, with algebraic multiplicity at least k , and the remaining $n - k$ eigenvalues are the spectrum of $\hat{B}^{-1}(T - W^{\top}V^{-1}W)$.

We examine this matrix block more closely. By the Sherman-Morrison-Woodbury formula $\hat{B}^{-1} = (\Phi - \tilde{W}^{\top}\tilde{V}^{-1}\tilde{W})^{-1} = (\Phi^{-1} + \Phi^{-1}W^{\top}(V - W\Phi^{-1}W^{\top})^{-1}W\Phi^{-1})^{-1}$, which we apply against $T - W^{\top}V^{-1}W$ to get

$$\begin{aligned} \hat{B}^{-1}(T - W^{\top}V^{-1}W) &= \Phi^{-1}T + \Phi^{-1}W^{\top}(V - W\Phi^{-1}W^{\top})^{-1}W\Phi^{-1}T \\ &\quad - \Phi^{-1}W^{\top}V^{-1}W - \Phi^{-1}W^{\top}(V - W\Phi^{-1}W^{\top})^{-1}W\Phi^{-1}W^{\top}V^{-1}W. \end{aligned}$$

We previously showed $\Phi^{-1}T = I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR}$, so now

$$\begin{aligned} \hat{B}^{-1}(T - W^{\top}V^{-1}W) &= I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR} + [\Phi^{-1}W^{\top}(V - W\Phi^{-1}W^{\top})^{-1}W(I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR}) \\ &\quad - \Phi^{-1}W^{\top}V^{-1}W - \Phi^{-1}W^{\top}(V - W\Phi^{-1}W^{\top})^{-1}W\Phi^{-1}W^{\top}V^{-1}W]. \end{aligned}$$

We set aside the terms $I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR} + (\Phi^{-1}W^{\top}(V - W\Phi^{-1}W^{\top})^{-1}W)(\Phi^{-1}T^{SN} + \Phi^{-1}T^{LR})$, and show the remaining terms cancel out.

$$\begin{aligned}
& \Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W - \Phi^{-1}W^\top V^{-1}W - \Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W\Phi^{-1}W^\top V^{-1}W \\
&= [-\Phi^{-1}W^\top V^{-1} + \Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}(I - W\Phi^{-1}W^\top V^{-1})]W \\
&= [-\Phi^{-1}W^\top V^{-1} + \Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}(V - W\Phi^{-1}W^\top)V^{-1}]W \\
&= [-\Phi^{-1}W^\top V^{-1} + \Phi^{-1}W^\top V^{-1}]W = 0.
\end{aligned}$$

So the clustering depends on the spectrum of $I + \Phi^{-1}T^{SN} + \Phi^{-1}T^{LR} + (\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)(\Phi^{-1}T^{SN} + \Phi^{-1}T^{LR})$. Most of the non-identity terms here are easily characterized using the previous results as either low-rank or small-norm. However, we must assume $(\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}T^{SN}$ is small-norm. Numerical validation of this assumption is found in Section (5.1). $\square$

We have now shown that M satisfies all three criteria we sought in a preconditioner, as set forth in Section (3.1).

5 Numerical Results

5.1 Validating Assumptions

In order to prove the spectral clustering, we made two assumptions about the structure of A : the blocks adjacent to the diagonal are well-approximated by low-rank matrices and the other off-diagonal blocks are small-norm. Table (1) displays the maximum and average (median) rank needed to achieve an approximation under ε on each block. Table (2) shows the norm and relative norm of T_F . Both tables include results for $\alpha = 1.3, 1.5, 1.9$, after 7 levels of refinement, but similar trends were seen across all fractional orders and refinement levels. In the case of 1×1 Toeplitz blocks we found that including the entry to the right and underneath in T_E was not sufficient to capture all the low-rank information. Instead, the corresponding block in T_E was extended to dimensions 5×5 such that the original entry is the center of that block. In terms of spectral clustering, this could increase the number of outliers, but better characterizes the size of the cluster.

5.2 Improved Convergence

Example results of the spectrum before and after preconditioning are in Figure (3). These tests are for the problem (1) with $\alpha = 1.7$ and exact solution $u(x) = 10x^2(1 - x^2)$ on $[0, 1]$ (see [17] example 6.1 for more details). The three matrix sizes are $n = 31, 71, 2795$ and reflect 0, 3, and 11 levels of refinement respectively. Note that zero levels of refinement is a uniform mesh.

The left-most column represents a matrix that consists of a single Toeplitz block, so the preconditioner reduces to the case of φ . In this case the clustering of the spectrum around 1 is very strong. In the $n = 71$ case, the biggest Toeplitz

$\varepsilon \backslash \alpha$	1.3	1.5	1.9
10^{-3}	4(5)	4(6)	5(6)
10^{-4}	5(7)	5(7)	6(8)
10^{-5}	6(8)	6(9)	7(9)

Table 1: The approximation rank needed for the error to be less than ε for varying fractional indices α . Table entries are displayed as **median (max)**.

α	1.3	1.5	1.9
$\ T_F\ _2$	0.6138	1.528	4.479
$\frac{\ T_F\ _2}{\ A\ _2}$	4.893×10^{-2}	2.368×10^{-2}	2.533×10^{-3}

Table 2: The norm of $\|T_F\|_2$ and $\|T_F\|_2/\|A\|_2$ across values of α .

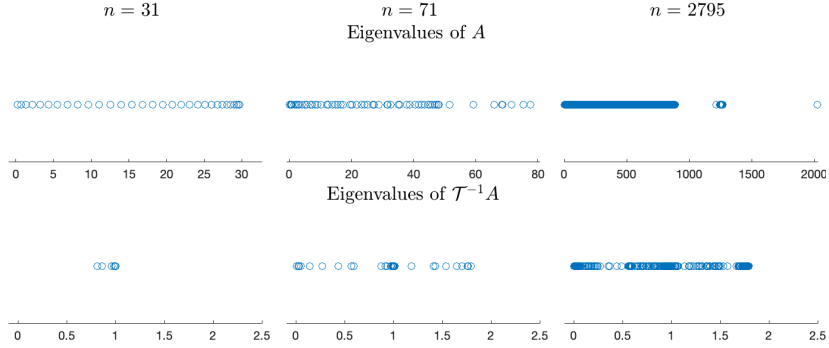


Figure 3: Eigenvalue clustering before and after preconditioning. Each column represent a different matrix size. The first row shows the spectrum before preconditioning, the second shows the spectrum, on a smaller interval, after preconditioning.

blocks are size 11, and there are two such blocks. The eigenvalue clustering around 1 is still observable, but the size of the cluster has grown. Finally for $n = 2795$, the preconditioner reduces the spectral radius by several orders of magnitude. The spectrum is concentrated around 1, but the distribution of eigenvalues between 0 and 2 is more uniform. For this matrix, the biggest Toeplitz block is size 939 and there are two other blocks just bigger than 200. On the other hand, there are 245 1×1 blocks. For all cases, the strength of the clustering is explained by the amount of smaller Toeplitz blocks in A . Notice that although the proof allows for eigenvalue outliers beyond the cluster, in practice everything remains within the interval $(0, 2)$.

Next, we see how the clustered spectrum translates to accelerated convergence. In Figure (4) the same matrices are used for preconditioned conjugate gradient (PCG), comparing the adaptive φ preconditioner and a Jacobi pre-

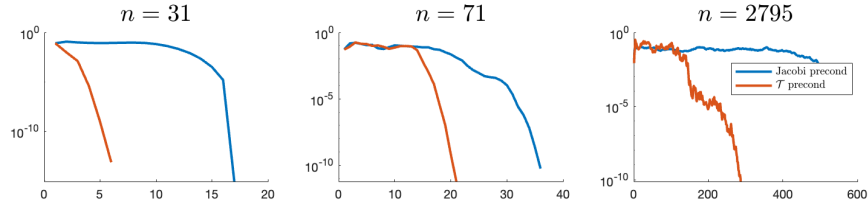


Figure 4: Comparing the residual vector norm across PCG iterations. The blue line uses a Jacobi preconditioner and the orange line uses the Φ preconditioner.

conditioner with a tolerance of 10^{-8} . For the two smaller matrices, the preconditioner clearly accelerates the point of superlinear convergence [14]. For the final matrix, the Jacobi-preconditioned system fails to converge within the maximum of 600 iterations, but with the Φ preconditioner converges in about 300 iterations.

6 Conclusion

We have shown how to build a preconditioner for FDEs discretized by AFEM. The spectrum of the preconditioned system is clustered around 1 and accelerates convergence of iterative methods. In the future, we will investigate how to customize the AFEM MARK module to build bigger Toeplitz blocks and thus get tighter clusters. Finally, we will investigate how to use these techniques to build preconditioners for FDEs discretized with AFEM on higher dimensional meshes.

References

- [1] DARIO BINI AND FABIO DI BENEDETTO, *A new preconditioner for the parallel solution of positive definite toeplitz systems*, Association for Computing Machinery, (1990), p. 220–223.
- [2] JAMES R. BUNCH, *Stability of methods for solving toeplitz systems of equations*, SIAM Journal on Scientific and Statistical Computing, 6 (1985), pp. 349–364.
- [3] RAYMOND H. CHAN AND MICHAEL K. NG, *Conjugate gradient methods for toeplitz systems*, SIAM Review, 38 (1996), pp. 427–482.
- [4] RAYMOND H. CHAN AND GILBERT STRANG, *Toeplitz equations by conjugate gradients with circulant preconditioner*, SIAM Journal on Scientific and Statistical Computing, 10 (1989), pp. 104–119.
- [5] RAYMOND H. CHAN AND MAN-CHUNG YEUNG, *Circulant preconditioners constructed from kernels*, SIAM Journal on Numerical Analysis, 29 (1992), pp. 1093–1103.
- [6] TONY F. CHAN, *An optimal circulant preconditioner for toeplitz systems*, SIAM Journal on Scientific and Statistical Computing, 9 (1988), pp. 766–771.
- [7] CHAO CHEN, XI YANG, AND FEI-YAN ZHANG, *Sine-transform-based fast solvers for riesz fractional nonlinear schrödinger equations with attractive nonlinearities*, Applied Mathematics and Computation, 510 (2026), p. 129674.
- [8] N. ENGHETA, *On fractional calculus and fractional multipoles in electromagnetism*, IEEE Transactions on Antennas and Propagation, 44 (1996), pp. 554–566.
- [9] BANGTI JIN AND ZHI ZHOU, *A Finite Element Method with Singularity Reconstruction for Fractional Boundary Value Problems*, ESAIM: Mathematical Modelling and Numerical Analysis, 49 (2015), pp. 1261–1283.
- [10] THOMAS KAILATH AND VADIM OLSHEVSKY, *Displacement structure approach to discrete-trigonometric-transform based preconditioners of g.strang type and of t.chan type*, SIAM Journal on Matrix Analysis and Applications, 26 (2006), pp. 706–734.
- [11] XIANJUAN LI, ZHIPING MAO, NAN WANG, FANGYING SONG, HONG WANG, AND GEORGE EM KARNIADAKIS, *A fast solver for spectral elements applied to fractional differential equations using hierarchical matrix approximation*, Computer Methods in Applied Mechanics and Engineering, 366 (2020).

- [12] RICARDO NOCHETTO, KUNIBERT SIEBERT, AND ANDREAS VEESER, *Theory of adaptive finite element methods: An introduction*, in Multiscale, Nonlinear and Adaptive Approximation, 2009, pp. 409–542.
- [13] MANUEL D. ORTIGUEIRA, CLARA M. IONESCU, J. TENREIRO MACHADO, AND JUAN J. TRUJILLO, *Fractional signal processing and applications*, Signal Processing, 107 (2015), p. 197.
- [14] VALERIA SIMONCINI AND DANIEL SZYLD, *On the superlinear convergence of minres*, (2013).
- [15] GILBERT STRANG, *A proposal for toeplitz matrix calculations*, Studies in Applied Mathematics, 74 (1986), pp. 171–176.
- [16] HONG WANG AND NING DU, *A superfast-preconditioned iterative method for steady-state space-fractional diffusion equations*, Journal of Computational Physics, 240 (2013), pp. 49–57.
- [17] XUAN ZHAO, XIAOZHE HU, WEI CAI, AND GEORGE EM KARNIADAKIS, *Adaptive finite element method for fractional differential equations using hierarchical matrices*, Computer Methods in Applied Mechanics and Engineering, 325 (2017), pp. 56–76.